The Use of Neutron Diffraction for In-Situ Analysis of Thermal Processing of Additively Manufactured Materials

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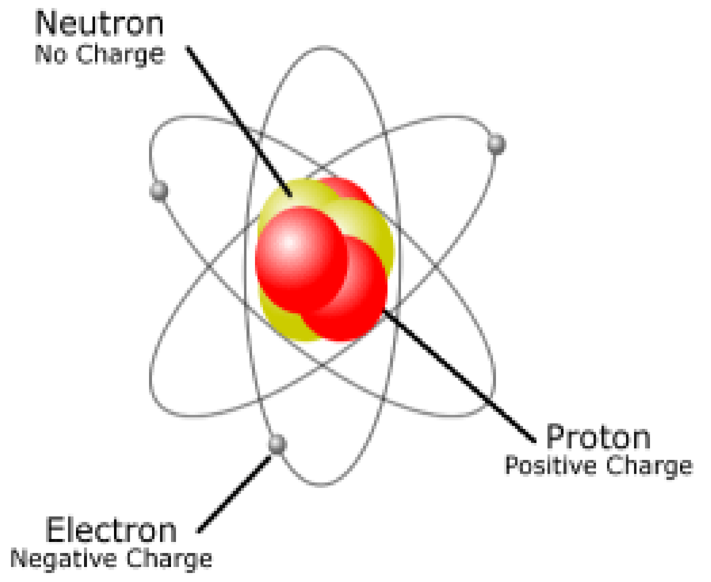
Introduction to Neutrons

- Neutrons:
  - Primary constituent of all atoms except hydrogen
  - Neutral in charge
  - Mass = 1.67495 x 10^{-27} kg
- Theorized in 1920 – Rutherford
- Proven in 1932 – Chadwick
- Generation of neutrons:
  - Fission or spallation reactions
  - Both take significant energy sources



ASM Metals Handbook 9th Ed. Vol. 10 Materials Characterization, ASM International 1986, pp 492 – 508.

ASM Metals Handbook 9th Ed. Vol. 10 Materials Characterization, ASM International 1986, pp 380 – 394.

https://byjus.com/chemistry/neutrons/ *(accessed 3/30/21)*

Introduction to Neutron Diffraction

- Neutron diffraction:
  - Evaluates: atomic structures, lattice parameters, residual stress, lattice strain, coefficient of thermal expansion, magnetic structure determination
  - Solid, powder, and liquid interrogation
  - High depth resolution (centimeters)
  - Long data acquisition times
    - 8 – 24 hours
  - Used in extreme environments
    - High pressure
    - High temperature

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Introduction to Additive Manufacturing

- <u>Additive Manufacturing (AM):</u>
  - Layer-by-layer build with powder or wire base material
  - Laser or electron beam melts subsequent layer
  - Program controls build geometry
  - Powder bed fusion (PBF) or directed energy deposition (DED)
- <u>Wire Arc Additive Manufacturing (WAAM):</u>
  - Wire-based AM where energy beam deposits wire onto molten pool on part surface.
  - Improved density and cheaper
- <u>AM Defects:</u>
  - Residual stresses from thermal cycles
  - Non-melted particles and porosity
  - Macroscopic/microscopic cracking
  - Microstructural inhomogeneity

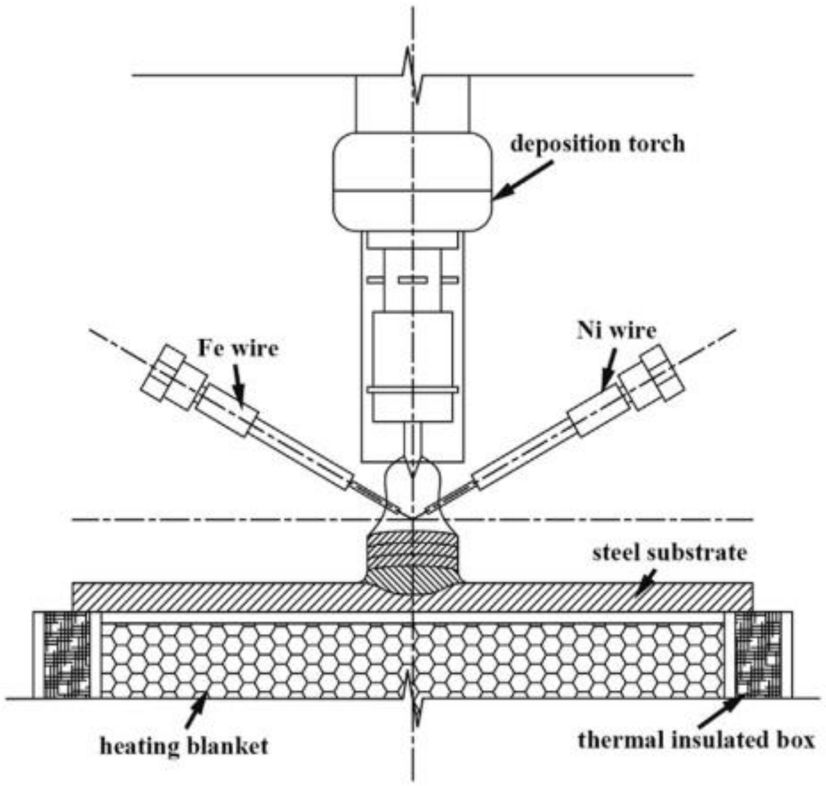

Illustration of a twin wire WAAM process, showing one Fe wire and one Ni wire. Wires are fed preferentially to generate the ideal compositional gradient required.

'Neutron Characterization for Additive Manufacturing', T. Watkins, H. Bilheux, K An, A. Payzant, R. Dehoff, C. Duty, W. Peter, C. Blue, C. Brice, Advanced Materials and Processes, 23 - 27 (2013).

'Thermal induced phase evolution of Fe-Fe3Ni functionally graded material fabricated using the wire-arc additive manufacturing process: An in-situ neutron diffraction study', C. Shen, X. Hua, M. Reid, K-D. Liss, G. Mou, Z. Pan, Y. Huang, H. Li, Journal of Alloys and Compounds **826** 154097 (2020).

Neutron Diffraction vs. X-Ray Diffraction

- <u>Neutron diffraction:</u>
  - Has higher sensitivity to lower atomic #
  - Determine magnetic information
  - Better depth resolution
  - Good contrast between neighboring elements

- <u>X-Ray diffraction:</u>
  - Short data acquisition time
  - Good spatial resolution
  - Variety of scattering strengths
  - More widely available tool
  - Sample does not become radioactive

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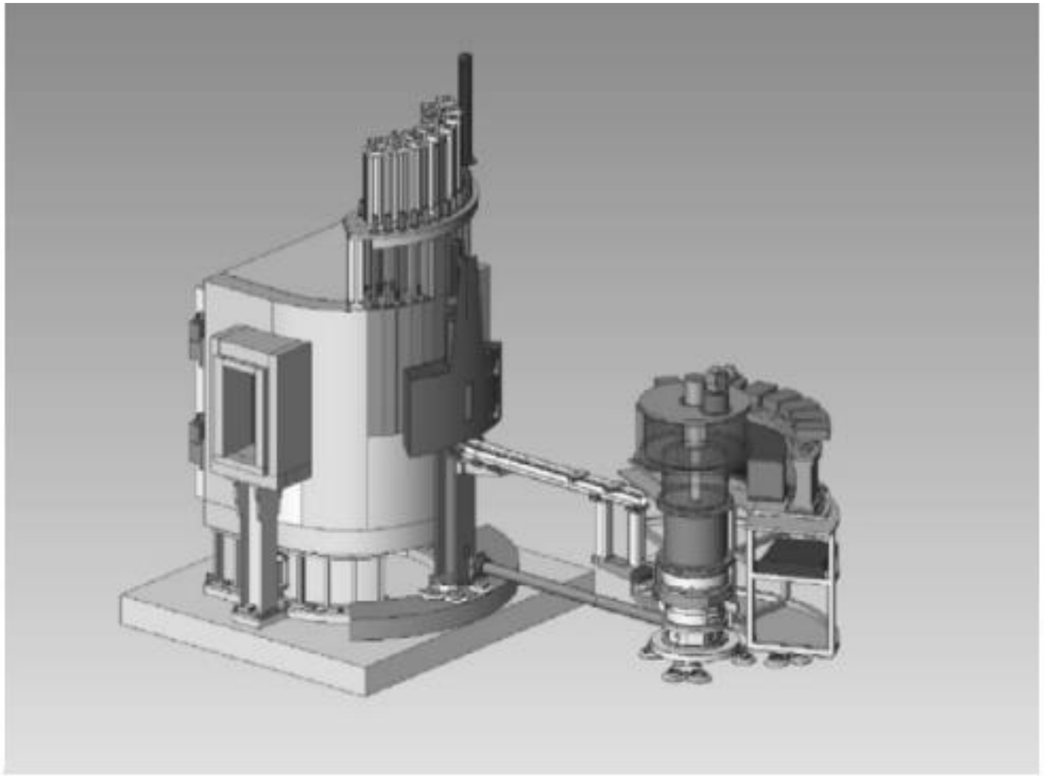
Functional Description of Neutron Diffraction

- Incident beam:
  - Neutrons that interrogate sample at specific wavelength that satisfies Bragg's Law
- Diffraction:
  - Probing neutrons enter sample and scattered beams are reinforced and the beam is strong enough to reach the detector.
  - Bragg's law: $\lambda = 2d_{hkl} * \sin\theta$,
    - λ is the incident wavelength of the neutrons
    - d is the interplanar lattice spacing
    - θ is the angle of the diffraction



WOMBAT instrumentation schematic with the sample table on the left and detector on the right

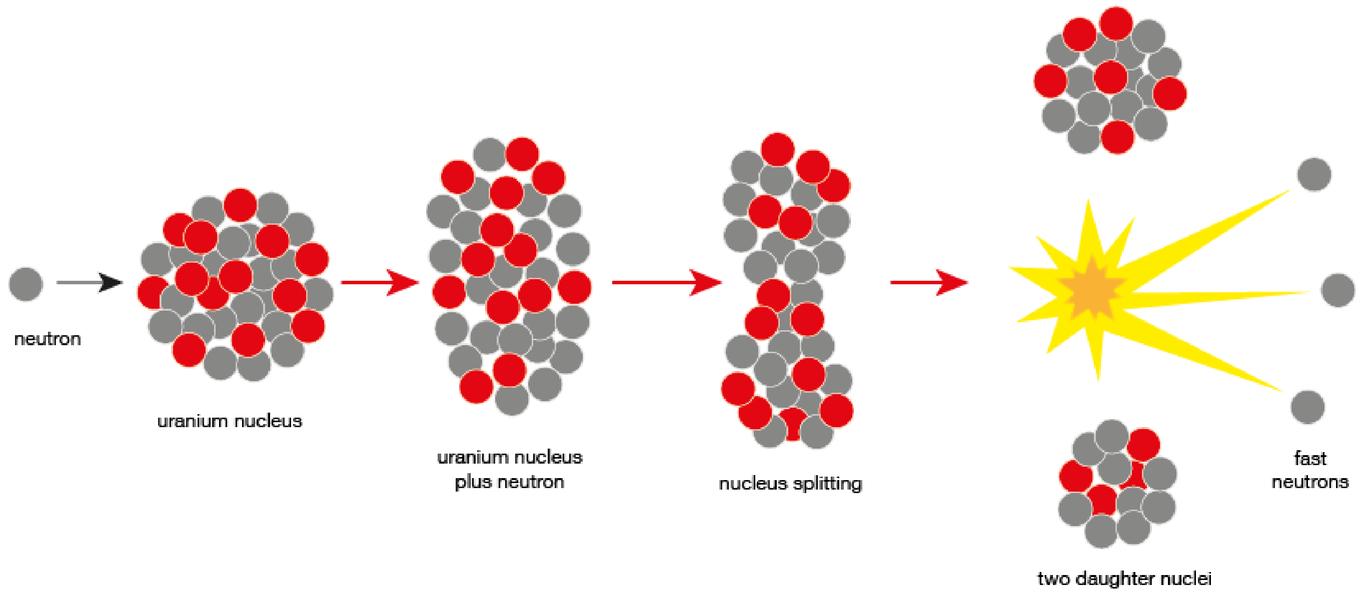
ASM Metals Handbook 9th Ed. Vol. 10 Materials Characterization, ASM International 1986, pp 380 – 394.

'Wombat: The high-intensity powder diffractometer at the OPAL reactor', A. Studer, M. Hagen, T. Noakes, Physica B **385-386**, 1013 – 1015 (2006).

Neutron Formation

- **Fission:**
  - Continuous source beam
  - Fission of ^{235}U or ^{239}Pu in nuclear reactors
    - Radioactive decay
    - Nucleus of atom splits
  - High flux reactors with ~14 MeV energy neutrons are generated

- **Spallation:**
  - Pulsed source beam
  - High energy protons impact W or Hg target, releasing neutrons
  - Source used in time-of-flight mode
  - Neutron release is t_0 for TOF



Fission chain reaction producing continuous source of neutrons

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https://mlz-garching.de/englisch/neutron-research/neutron-source.html *(accessed 3/30/21)*

Neutron Filtering

- Monochromatic single crystal filters neutron beam that utilizes desired Bragg peak to filter and steer neutron beam.
  - Generally utilized with fission neutron sources.
- Time-of-flight beam filtering sorts neutrons by energy by measuring the time between generation and arrival to detector.
  - Used with spallation neutron sources.

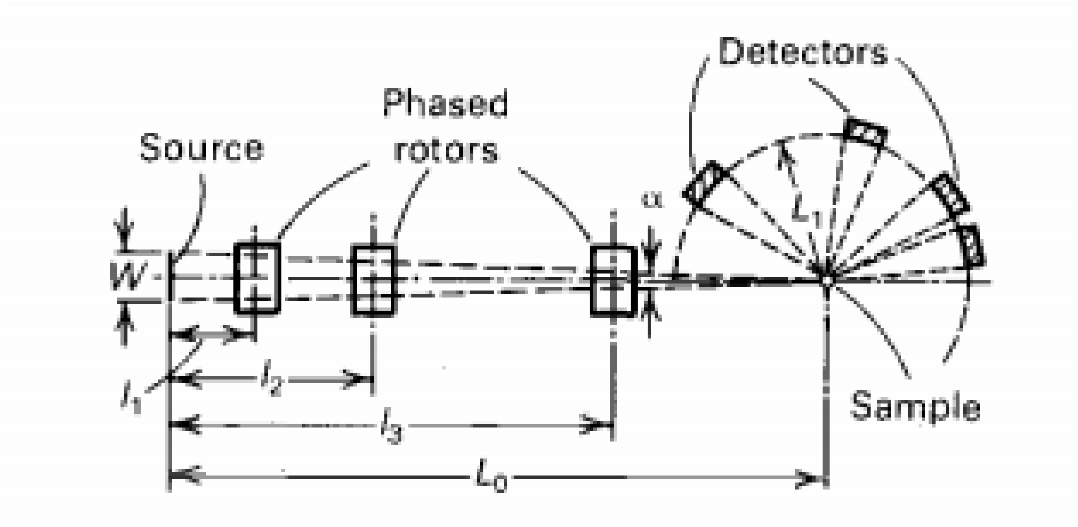
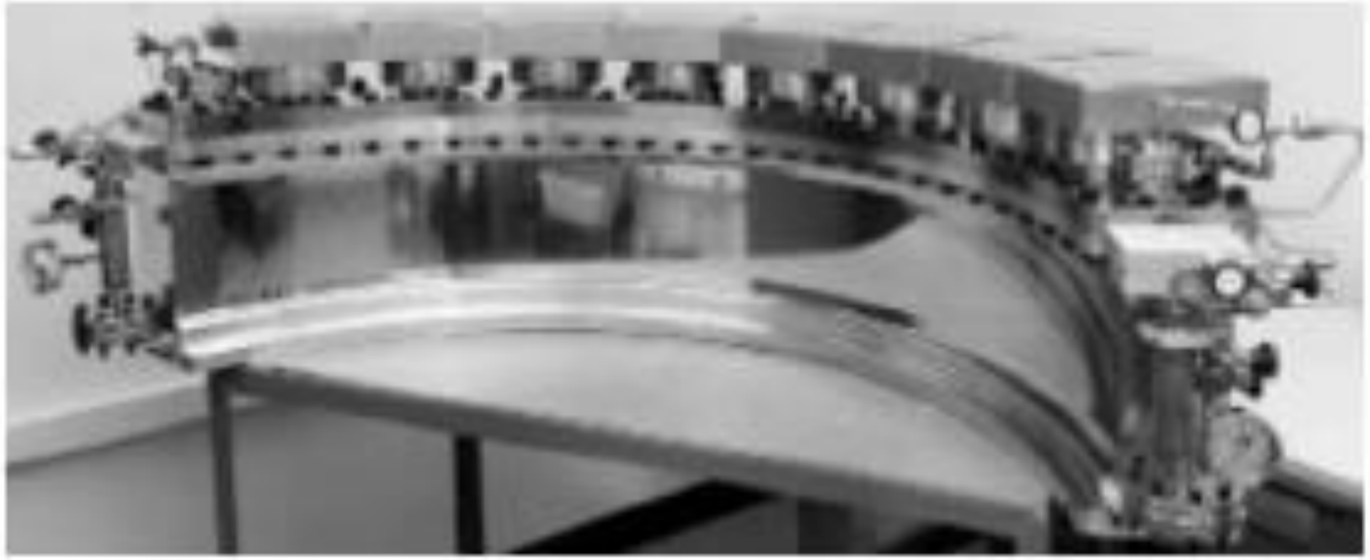


Diagram of a time-of-flight powder diffractometer which measures incident wavelength of the neutrons based on travel time from source

ASM Metals Handbook 9th Ed. Vol. 10 Materials Characterization, ASM International 1986, pp 380 – 427.

Neutron Detection

- Neutron detectors are bulky and expensive.
- Designed specific to source and filtering system.
- Helium-3 gas filled proportional counter typically used for detection of neutrons.



Detector used in conjunction with WOMBAT system shown on Slide 6.

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'Wombat: The high-intensity powder diffractometer at the OPAL reactor', A. Studer, M. Hagen, T. Noakes, Physica B **385-386**, 1013 – 1015 (2006).

Case Study 1

- Functionally Graded Material (FGM) Fe-Fe_3Ni samples produced via WAAM

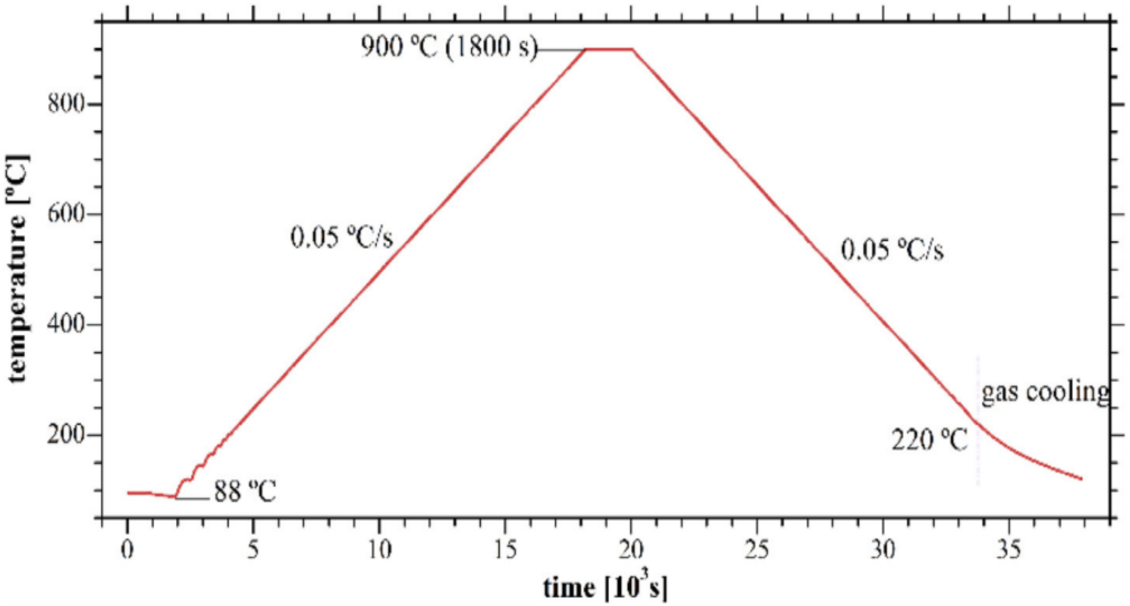
- WAAM FGM has susceptibility to macroscopic cracking
  - Alpha ferrite leads to cold cracking
  - A post-build heat treatment is utilized to homogenize the residual bcc α-ferrite phase into the fcc Fe-Fe_3Ni FGM matrix

- In-situ neutron diffraction used to:
  - Quantify the phase transformation during a post-build homogenization thermal treatment
  - Measure the thermal expansion during the heat treatment and determine its effects on the hot-cracking phenomenon

'Thermal induced phase evolution of Fe-Fe3Ni functionally graded material fabricated using the wire-arc additive manufacturing process: An in-situ neutron diffraction study', C. Shen, X. Hua, M. Reid, K-D. Liss, G. Mou, Z. Pan, Y. Huang, H. Li, Journal of Alloys and Compounds **826** 154097 (2020).

Case Study 1 – Experimental Set Up

- High-flux WOMBAT diffractometer used in conjunction with a TG1 thermal neutron guide at the OPAL reactor
- Vertically focusing Ge 115 monochromator uses 23 crystal plate
- 2D position detector allows for analysis in 120^{o} in 2θ for which the major sensitive area is set up for high-speed recording
- Detector filled with 7atm ^{3}He and 2.5 atm propane



Thermal processing schedule used for post-build Fe-Fe_3Ni FGM samples for the in-situ neutron diffraction study.

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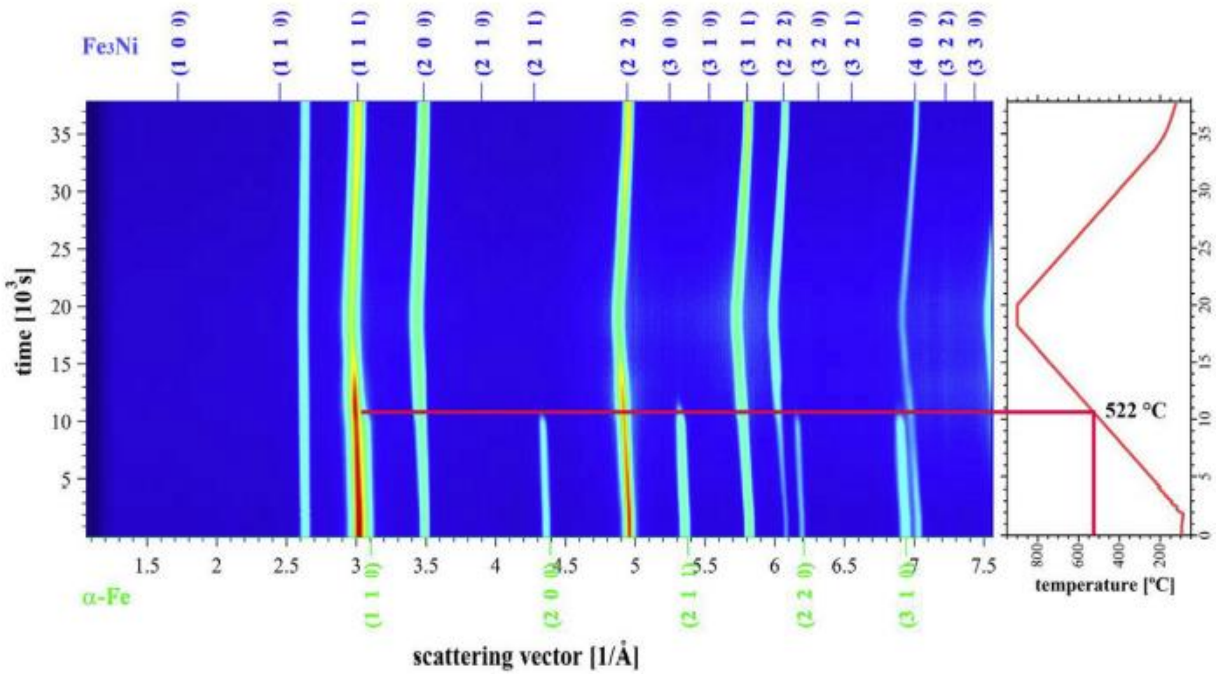
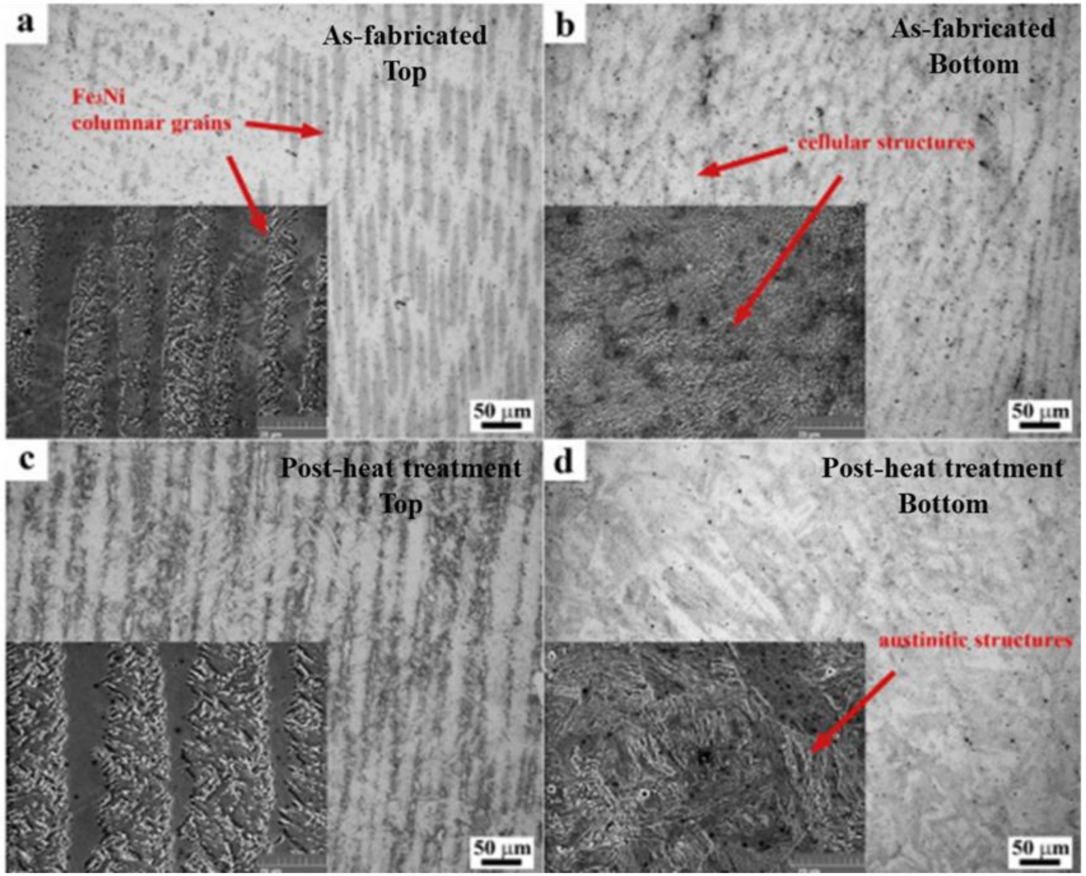
Case Study 1 – Results





- Neutron diffractograms confirm the results of the microscopy in that the α-Fe phase disappears at 522°C, and is transformed into an *fcc* structure.
- α-Fe phase is dissolved as no evidence of this phase is shown in the scattering pattern post thermal treatment.

- Microstructural of the top (left) and bottom (right) of the WAAM samples both as-fabricated (top) and post thermal processing (bottom).
- Microstructural analysis confirmed dissolution of ferritic cellular structures in the bottom of the build and growth of the columnar grain structure in the top of the build

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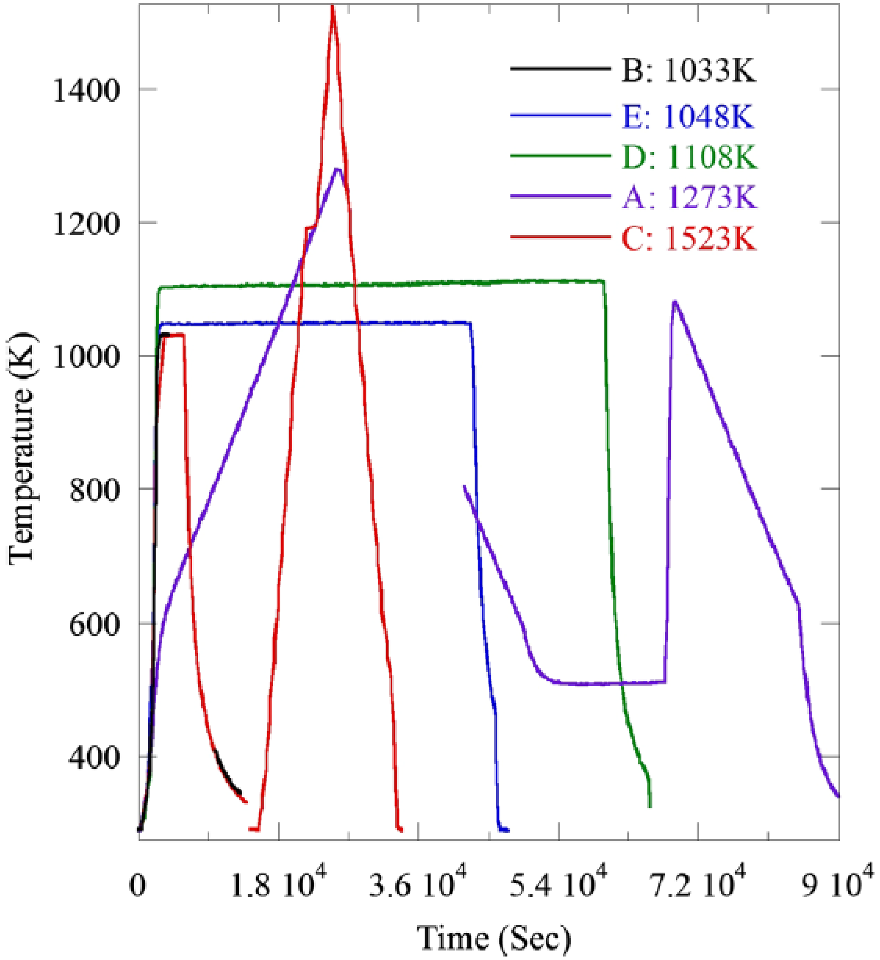
Case Study II

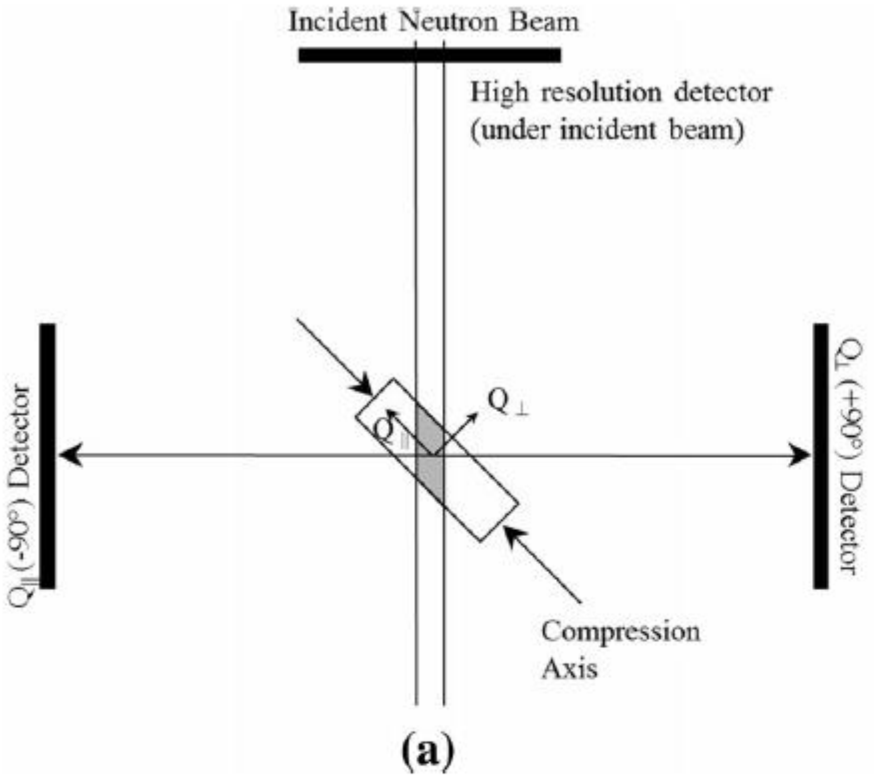
- DED manufacturing of 304L SS (stainless steel) samples.
  - Rapid cooling and large thermal gradients lead to dislocations.
- Samples electrical discharge machined from single build.
- In-situ neutron diffraction used to:
  - Evaluate the effects of the additively manufactured dislocation density, ferrite content, chemical heterogeneity, and grain morphology on macroscopic strength



Temperature exposure for each sample that was analyzed

'Using In-Situ Neutron Diffraction to Isolate Specific Features of Additively Manufactured Microstructures in 304L Stainless Steel and Identify Their Effects on Macroscopic Strength' D.W. Brown, D. P. Adams, L. Balogh, J.S. Carpenter, B. Clausen, V. Livescu, R.M. Martinez, B.M. Morrow, T.A. Palmer, R. Pokharel, M. Strantza, S.C. Vogel, Metallurgical and Materials Transactions A: Physical Metallurgy and Materials Science, **50**, 3399-3413 (2019)

Case Study II – Experimental Set Up



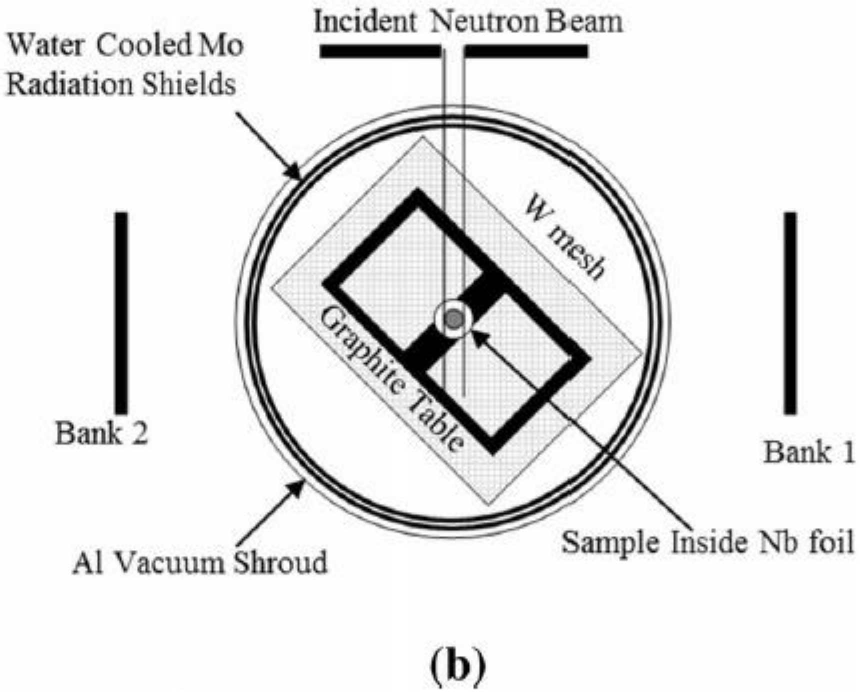


Illustration of the SMARTS sample set up with:

a) Scattering vectors and compression axis and that was used to measure lattice strain under load/compression

b) Setup used to conduct neutron diffraction measurements of phase transformation during thermal processing

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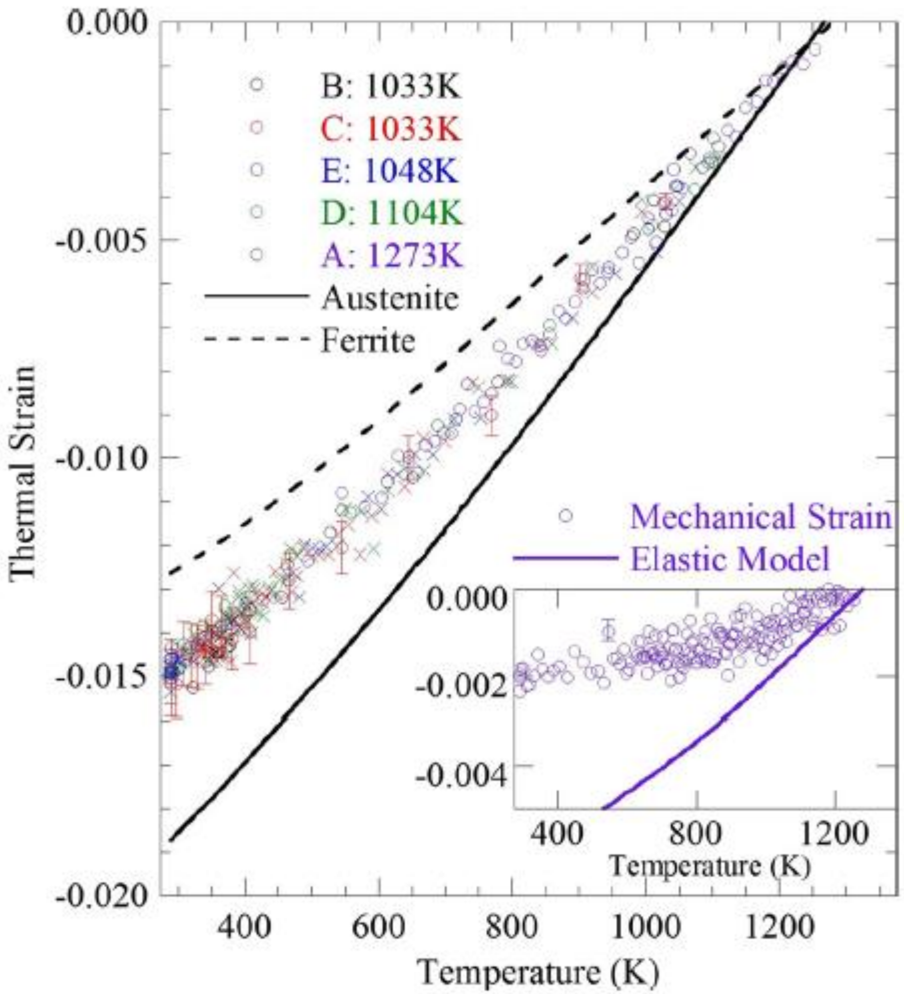
Case Study II – Results

- Dislocation density was reduced as a function of heat exposure.
  - Dislocation dissolution was the first microstructural change to occur during thermal processing
- At 1523 K, the microstructure recrystallized and grain growth was observed in conjunction with a reduction of ferritic phase volume



Thermal strains in the ferritic phases during the thermal processing. The inset shows predicted strain (solid line) versus the measured (circles). The dashed line is the CTE of ferrite.

‘Using In-Situ Neutron Diffraction to Isolate Specific Features of Additively Manufactured Microstructures in 304L Stainless Steel and Identify Their Effects on Macroscopic Strength’ D.W. Brown, D. P. Adams, L. Balogh, J.S. Carpenter, B. Clausen, V. Livescu, R.M. Martinez, B.M. Morrow, T.A. Palmer, R. Pokharel, M. Strantza, S.C. Vogel, Metallurgical and Materials Transactions A: Physical Metallurgy and Materials Science, **50**, 3399-3413 (2019)

Summary

- Neutron diffraction allows for in-situ non-subjective analysis of multiphase materials in a variety of thermal and pressure environments.

- The depth resolution and allowance to measure changes in the volume of a sample make this material characterization method ideal for in-situ thermal processing evaluation of additively manufactured materials.

- The use of neutron diffraction has and will continue to allow for post-fabrication thermal process optimization in order to achieve desired microstructure and internal stress states in additively manufactured parts.

Thank you!